Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	9th march 2026
Team ID	EL2026TMID10104
Project Name	AI-Based Anomaly Detection Platform
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through secure web form. Users can register using institutional email, Gmail, or LinkedIn authentication.
FR-2	User Confirmation	System verifies the user through email verification link or OTP before allowing access to the dashboard.
FR-3	Real-Time Vital Data Streaming	System collects patient vitals such as heart rate, SpO_2, body temperature, and blood pressure and streams the data using Apache Kafka for continuous monitoring.
FR-4	Anomaly Detection	Machine learning models (Autoencoder Neural Network and Isolation Forest) analyze incoming health data to detect abnormal patterns and compute anomaly scores.
FR-5	Severity Classification	Detected anomalies are categorized into severity levels such as LOW, MEDIUM, and HIGH based on anomaly scores.
FR-6	Data Storage	All patient vitals and anomaly results are stored securely in a PostgreSQL database for further analysis and record keeping.
FR-7	Dashboard Visualization	A Flask-based interactive dashboard displays real-time patient vitals, anomaly alerts, and historical health trends.
FR-8	Alert Notification System	The system automatically sends email notifications to healthcare administrators when a HIGH severity anomaly is detected.

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The system should provide an intuitive and easy-to-use dashboard for healthcare staff to monitor patient health conditions efficiently.
NFR-2	Security	The system must ensure secure authentication and protect sensitive healthcare data using encryption and secure communication protocols.
NFR-3	Reliability	The system should consistently process real-time health data streams without data loss and ensure accurate anomaly detection.
NFR-4	Performance	The platform must process incoming patient data with minimal latency so that anomalies can be detected and alerts generated in real time.
NFR-5	Availability	The monitoring system should maintain high uptime to ensure continuous health monitoring and timely alerts for critical situations.
NFR-6	Scalability	The system must be capable of handling an increasing number of patients and data streams using scalable technologies like Apache Kafka and distributed systems.